

Project Design Phase-II Technology Stack (Architecture & Stack)

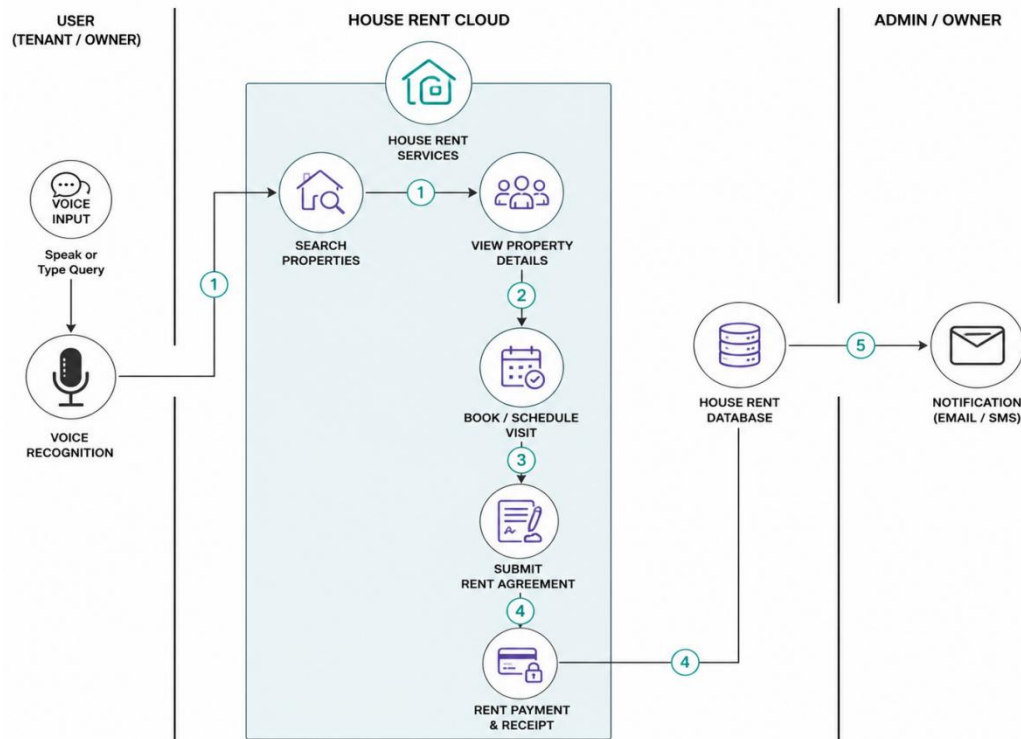
Date	13 july 2026
Team ID	SWTID-2026-7371
Project Name	House Rent
Maximum Marks	4 Marks

Technical Architecture:

The technical architecture illustrates the components and workflow of the HouseHunt - House Rental Management System.

Example: HouseHunt – House Rental Management System

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- **Include application processes:** User Registration, Login, Property Search, Property Listing, Booking, Admin Approval, and Dashboard.
- **Provide infrastructure:** Separate Frontend (React), Backend (Node.js & Express.js), MongoDB, and Cloud services.
- **Indicate external interfaces:** Google Maps API, JWT Authentication, Email/SMS Notifications, and Payment Gateway (if used).
- **Indicate data storage:** Store Users, Properties, Bookings, Images, and Rental Details in MongoDB.
- **Machine Learning (Optional):** Property Recommendations or Rent Price Prediction.

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How users interact with the application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / React.js
2.	Frontend Framework	Framework used to build responsive and interactive user interface	React.js
3.	Backend Logic	Handles business logic, user requests, and server-side operations	Node.js / Express.js
4.	Authentication & Authorization	User registration, login, role-based access for tenants, owners and admin	JWT (JSON Web Token), bcrypt.js
5.	Database	Stores application data such as users, properties, bookings, reviews etc.	MongoDB
6.	Cloud Database Service	Database service hosted on cloud	MongoDB Atlas
7.	File Storage	Stores property images, documents and other uploaded files	Cloudinary / Firebase Storage
8.	External API - Maps	Used for location search, distance, and map integration	Google Maps API
9.	Email Service	Used for sending verification emails, booking confirmations, notifications	Nodemailer / SendGrid
10.	Payment Gateway	Handles secure online payments for bookings (if applicable)	Razorpay / Stripe
11.	Infrastructure (Server / Cloud)	Application deployment on cloud platform for high availability	AWS / Render / Vercel / Heroku

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used in the application	React.js, Node.js, Express.js
2.	Security Implementations	List all the security / access controls implemented, use of firewalls, encryption etc.	HTTPS, JWT Authentication, Password Hashing, Input Validation, Helmet.js, CORS, Rate Limiting
3.	Performance Optimization	Mechanisms used to improve the performance and response time of the application	Code Splitting, Image Optimization, Caching, Database Indexing
4.	Scalability & Deployment	Approach used to scale the application and deployment strategy	Cloud Deployment, Load Balancing, Auto Scaling, CI/CD Pipeline